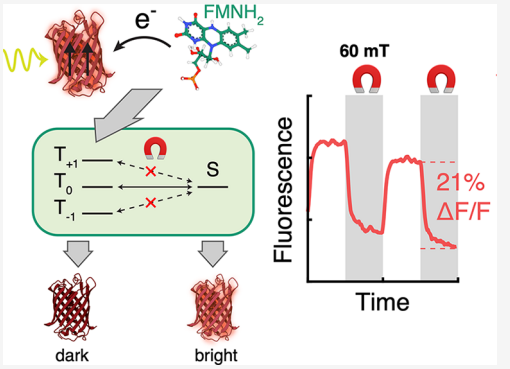
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Mechanism of Giant Magnetic Field Effect in a Red Fluorescent Protein

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ABSTRACT: Several fluorescent proteins, when expressed in *E. coli*, are sensitive to weak magnetic fields. We found that mScarlet3 fluorescence in *E. coli* reversibly decreased by 21% in the presence of a 60 mT magnetic field, the largest magnetic field effect (MFE) reported in any fluorescent protein. Purified mScarlet3 did not show an MFE, but the addition of flavin mononucleotide (FMN) and simultaneous illumination with blue and yellow light restored the MFE. Through extensive photophysical experiments, we developed a quantitative model of the giant MFE in mScarlet3-FMN mixtures. The key reaction step involved electron transfer from fully reduced $FMNH_2$ to triplet-state mScarlet3 to form a triplet spin-correlated radical pair. The magnetic field then controlled the branching ratio between singlet recombination vs triplet separation. Our quantitative model of the mScarlet3-FMN photocycle provides a framework for the design and optimization of magnetic-field-sensitive proteins, opening possibilities in fluorescent protein-based magnetometry, magnetic imaging, and magnetogenetic control.



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INTRODUCTION

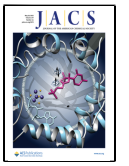
Interactions between milli-Tesla magnetic fields and organic matter at room temperature are too weak to affect reaction outcomes through thermodynamic means. Nonetheless, magnetic field effects (MFEs) on photochemical reactions have long been observed in organic dye solutions[2−5] and in some biochemical systems, but such effects are typically small (<1%)[6,7] or restricted to nonphysiological conditions.[8,9] Identification of large (>10%) MFEs in biochemical reactions under physiological conditions would open the possibility to harness these effects to modulate biochemical signals, or to implement quantum coherent measurement or control protocols in biology, and may shed light on possible mechanisms of magnetosensation.

A recent report described MFEs in several fluorescent proteins in live *E. coli*, and when paired with redox cofactors in vitro.[1] Enhanced green fluorescent protein (EGFP) fused to a flavin-binding tag showed a 0.25% MFE in *E. coli* when exposed to a 25 mT field. A mixture of purified mScarlet, a red fluorescent protein, and FMN showed a nearly 3% MFE in a 10 mT magnetic field, provided that the sample had been preilluminated with 470 nm light. Mutagenesis and screening in a light-oxygen-voltage-sensing (LOV) domain protein led to an astonishing 35% magnetic modulation in flavin fluorescence in an engineered variant termed magLOV.[1] The mechanisms underlying these newly discovered MFEs are not known.

MFEs are thought to arise through quantum dynamics in a spin-correlated radical pair intermediate (reviewed extensively in ref 10). Briefly, an electronic excited state — usually produced by photoexcitation — undergoes electron transfer to produce a pair of electrons that are weakly interacting but whose spin states are in a nonthermal distribution, favoring either singlet or triplet states, i.e. a spin-correlated radical pair (SCRP). Each electron spin precesses under the influence of distinct hyperfine interactions with nearby nuclear spins, leading to interconversion between singlet and triplet states, known as intersystem crossing (ISC). An external magnetic field partially decouples the electron and nuclear spins, suppressing ISC. Singlet radical pairs are more likely to recombine, while triplets are more likely to separate, leading to an MFE on the photochemical reaction outcome.

While the broad outlines of the SCRP mechanism are well established, the molecular species, reaction steps, and rate constants were not known for the MFE in any fluorescent protein. In particular, the reported requirement for preillumination with blue light in the mScarlet/FMN MFE was a

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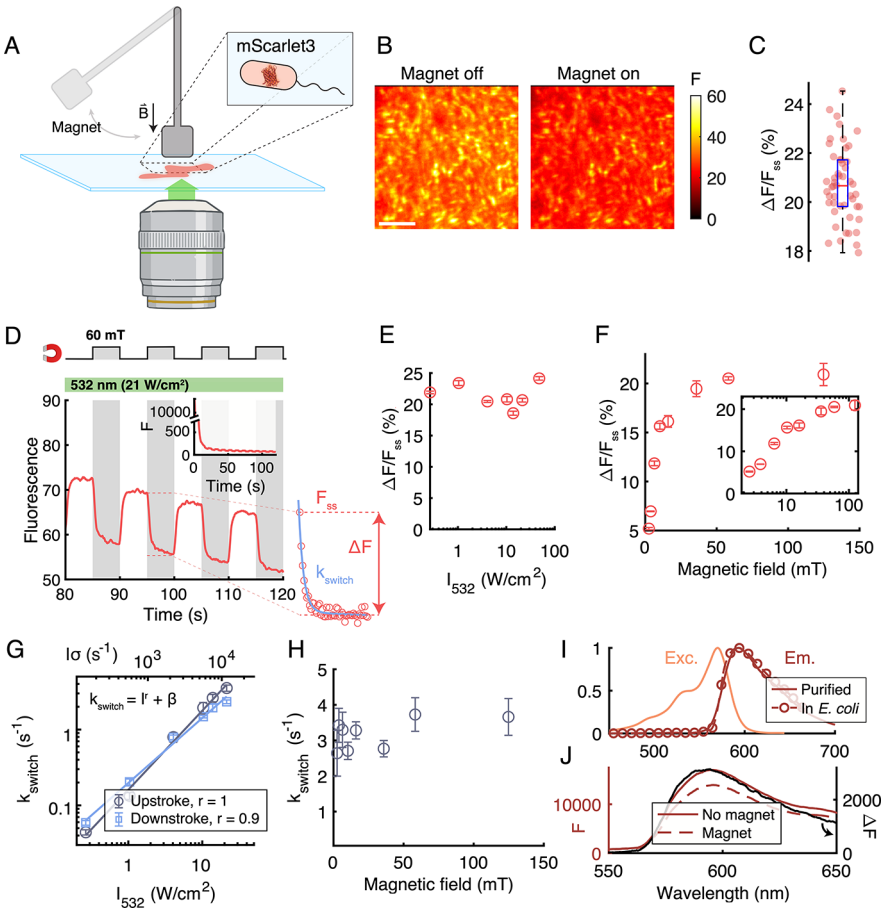


Figure 1. mScarlet3 fluorescence in *E. coli* is magnetic field sensitive. (A) Experimental setup. Fluorescence of *E. coli* expressing mScarlet3 (PDB ID: 7zct)[49−51] was probed via epifluorescence microscopy, while a magnet on a servo modulated the local magnetic field. (B) Image of *E. coli* expressing mScarlet3 in the absence (left) or presence (right) of a 60 mT magnetic field. Same contrast for both images. Scale bar 10 μm. (C) Distribution of amplitudes of the MFE across individual *E. coli*. $\Delta F/F_{ss}$ = (20.7 ± 1.5)% (mean ± s.d., n = 50 cells). Boxplot shows median, 25th and 75th percentiles; whiskers show extrema. (D) Fluorescence time-trace of mScarlet3 under green excitation (532 nm, 21 W/cm^2) and alternating magnetic field. Inset: upon illumination onset, fluorescence rapidly quenched. Right: definitions of the parameters k_{switch}, ΔF, and F_{ss}. (E) Steady state MFE amplitude as a function of illumination intensity. Magnetic field was 60 mT. (F) Steady state MFE amplitude as a function of magnetic field. Illumination intensity was 21 W/cm^2. (G) Switching rate as a function of illumination intensity. Magnetic field was 60 mT. Rates were approximately first-order in illumination intensity for both the upstroke and the downstroke. (H) MFE switching rate for the upstroke as a function of magnetic field. Illumination intensity was 21 W/cm^2. Rates were independent of magnetic field. (I) Excitation and emission spectra of purified mScarlet3 (solid curve), overlaid with emission spectrum of *E. coli* expressing mScarlet3 (circles). (J) Magnetic field modulated the amplitude (red traces), but not the shape (black trace), of the fluorescence emission spectrum. Error bars for E−H denote s.e.m.

mystery. In a previous report on an MFE involving flavin and proteins, the photoexcited flavin oxidized the protein.[7] In the mScarlet3/FMN system, we find that photochemically produced $FMNH_2$ reduces the protein. The MFE photocycle provides a guide for efforts to apply or enhance protein-based magnetic field effects and for coherent control of spin dynamics in proteins.

RESULTS

mScarlet3 Shows a Large MFE in Live*E. coli*. We imaged the fluorescence of several red fluorescent proteins expressed in live *E. coli* using an epifluorescence microscope. We used a servo-controlled permanent magnet on a stick to switch the magnetic field between 0 and 60 mT (Figure 1A). We tested red-emitting fluorescent proteins mScarlet3, mScarlet-I3, mSandy2, mRuby3, and mKate2. All except mKate2 showed magnetically modulated fluorescence (Figure S1).

We focused on mScarlet3 because it was bright and had the largest MFE of the proteins we tested. In the presence of the magnetic field (B = 60 mT), the fluorescence of mScarlet3 was visibly dimmer than in its absence (Figure 1B; 532 nm excitation, 48 W/cm^2, 575 nm long-pass emission, B switching every 5 s). We quantified the MFE amplitude by $\Delta F/F_{ss}$, where F_{ss} is the steady-state fluorescence at magnetic field B = 0, and $\Delta F = F_{ss} - F(B)$. Quantification across individual mScarlet3-expressing bacteria showed an MFE of $\Delta F/F_{ss}$ = 20.7 ± 1.5% (mean ± s.d., n = 50 cells, Figure 1C).

We then explored how the illumination intensity and the magnetic field strength affected the dynamics of the fluorescence. Upon onset of illumination at 21 W/cm^2, the fluorescence rapidly quenched from 13,000 to 80 counts, with a time-constant of ∼1.8 s (Figure 1D inset). After the fluorescence stabilized, stepwise changes in magnetic field induced gradual changes in fluorescence. The steady-state MFE amplitude was independent of the excitation intensity over the measured range I = 0.3−48 W/cm^2 (Figures 1E and

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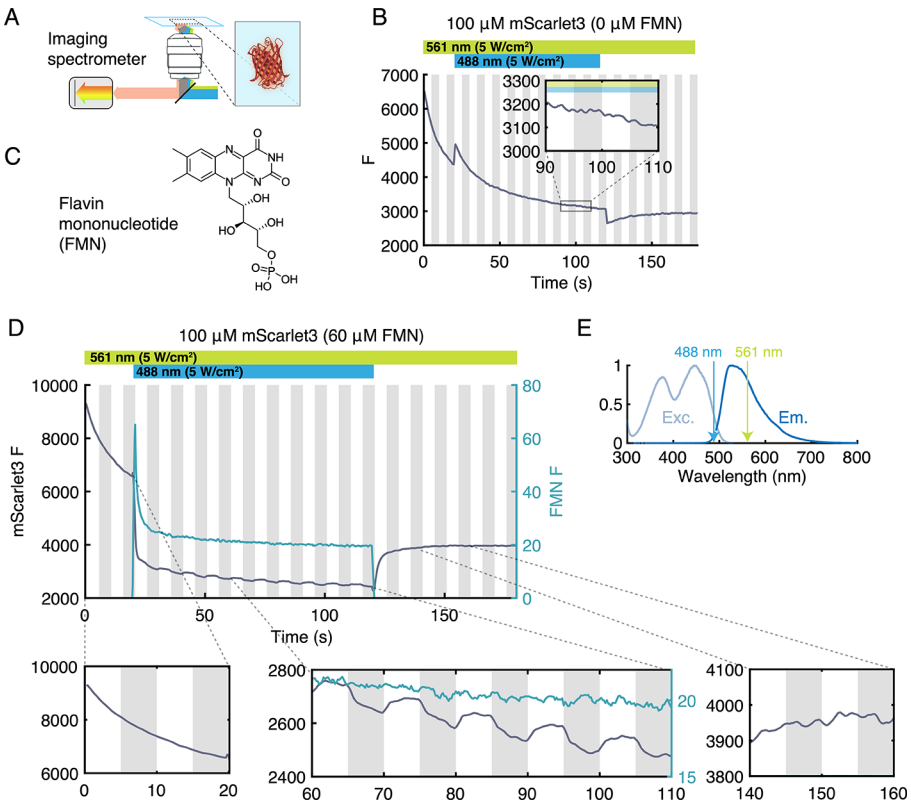


Figure 2. The mScarlet3 MFE requires FMN and blue light. (A) Fluorescence spectra were measured in a microscope with an imaging spectrometer. (B) Fluorescence of purified mScarlet3 showed no MFE under yellow (561 nm) or yellow + blue (488 nm) illumination. The magnetic field was modulated in 5 s intervals between B = 60 mT (gray stripes) and B = 0 (white stripes). (C) Chemical structure of FMN. (D) Addition of FMN (60 μM) did not affect mScarlet3 fluorescence under yellow-only illumination. Onset of blue illumination led to partial quenching of mScarlet3 fluorescence and appearance of an MFE. Fluorescence was spectrally filtered to separately record mScarlet3 and flavin emission. (E) Excitation and fluorescence emission spectra of FMN.

S2). The steady-state MFE amplitude increased with increasing magnetic field, with a half-saturation field $B_{1/2}$ = 5.5 mT (Figure 1F).

We fit the magnetic field-induced switching rate constant, k_{switch} (defined in Figure 1D), over an intensity range from I = 0.3−21 W/cm^2 (corresponding to per-molecule excitation rates $I\sigma$ = 145−11,600 s^{-1}; σ is the absorption cross-section at 532 nm and is proportional to ε_{532} = 53,040 M^{-1} cm^{-1}). The switching rate at magnetic field onset varied from 0.04 to 3.5 s^{-1} over this range, and was approximately linear in intensity, i.e., $k_{switch} \propto I^r + \beta$ with r = 1.0 for the fluorescence upstroke and r = 0.9 for the downstroke (Figure 1G). These results suggested a single rate-limiting photochemical step in the magnetic-field dependent photochemical reaction. The ratio of k_{switch} (s^{-1}) to per-molecule excitation rate $I\sigma$ (s^{-1}) gives a branching ratio of 3.1 × 10^{-4} for magnetic field-dependent photoswitching from bright to dark forms of mScarlet3.

The switching rate was independent of the magnetic field strength over the range 3−125 mT (Figure 1H), implying that the magnetically modulated rate constant(s) were not the rate-limiting steps in the photochemical conversion between fluorescent and non-fluorescent mScarlet3 states. The fluorescence emission spectrum of the *E. coli* closely matched the spectrum for purified mScarlet3 (Figure 1I),[11] and the MFE modulated the amplitude, but not the shape, of this spectrum (Figure 1J). These observations indicate that mScarlet3 was the primary emissive species.

In *E. coli*, various factors could influence the MFE including pH, redox factors, and diffusional confinement of molecules within each bacterium. We found that the MFE was largest on the second day of expression in liquid culture, suggesting the involvement of metabolic or redox factors beyond just the fluorescent protein. To study the molecular mechanisms underlying the MFE under controlled conditions, we thus switched to working with purified protein.

Purified mScarlet3 Shows an MFE in the Presence of FMN and Blue Light. We purified mScarlet3 and measured its fluorescence while varying the magnetic field. An imaging spectrometer recorded both the amplitude and shape of the emission spectrum (Figure 2A). No MFE was detected for yellow (561 nm) or yellow + blue (561 + 488 nm) excitation (Figure 2B). Prior work showed that the addition of flavin mononucleotide (FMN, Figure 2C) and blue light restored the MFE in mScarlet3,[1] so we replicated that result. Addition of 60 μM FMN did not produce an MFE under yellow light alone. However, when we combined blue and yellow light in the presence of FMN, we observed a rapid initial drop in mScarlet3 fluorescence, and then an MFE (Figure 2D). The blue light also evoked FMN fluorescence, which was spectrally distinct from mScarlet3 fluorescence. The FMN fluorescence showed a rapid initial drop before stabilizing, but did not show a detectable MFE. Upon turning off the blue light, the mScarlet3 fluorescence partially recovered, and the MFE went away.

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Based on these measurements, and the overlap of the blue laser wavelength with the FMN absorption spectrum (Figure 2E), we speculated that a photochemical product of FMN excitation might be involved in the MFE. To test this hypothesis, we prepared a sample comprising mScarlet3 immobilized on Ni-NTA resin beads, immersed in a solution of FMN (500 μM), and sandwiched between glass coverslips. We targeted the yellow light to focus on a single bead, and the blue light to pass through a region of FMN solution approximately 200 μm from the bead (Figure 3A,B). This

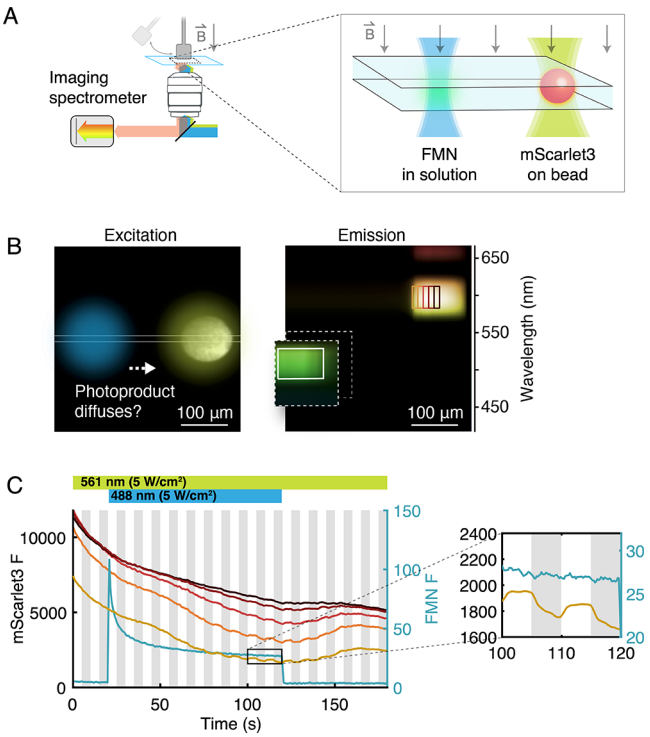


Figure 3. The mScarlet3 MFE requires a long-lived FMN photoproduct. (A) Experiment to test whether an FMN photoproduct contributed to the MFE. A bead containing tethered mScarlet3 was illuminated with yellow light. Blue light was laterally displaced and focused into a solution of FMN. (B) Left: excitation profiles overlaid on an image of the bead. Right: emission of FMN (green) and mScarlet3 (yellow) on an imaging spectrometer. The x-axis corresponds to position and the y-axis to emission wavelength. The contrast of the FMN emission has been increased to render it visible. (C) Time-course of FMN fluorescence (cyan) and mScarlet3 fluorescence in the regions on the bead indicated with correspondingly colored rectangles in (B). The magnetic field was modulated in 5 s intervals between B = 60 mT (gray stripes) and B = 0. Approximately 35 s after blue light onset, the mScarlet3 fluorescence showed quenching and onset of an MFE, with the regions closer to the blue light responding first. The FMN fluorescence was not detectably magnetic field sensitive.

arrangement prevented any direct blue light interaction with mScarlet3 molecules, while allowing diffusive interchange of long-lived photoproducts between the two illuminated regions. We measured the emission from both the blue-illuminated and yellow-illuminated regions with an imaging spectrometer (Figure 3B, right), while we varied the magnetic field.

As with the soluble proteins, the mScarlet3 initially showed no MFE under yellow-only illumination. When the blue light was turned on, the FMN fluorescence decayed over ~15 s, suggesting photochemical consumption of the FMN. As before, there was no detectable MFE in the FMN fluorescence. After a delay of ~35 s, the mScarlet3 fluorescence began to drop, and an MFE appeared (Figure 3C). The side of the bead closest to the blue spot had the earliest and largest response, and more distant regions had later and smaller responses. When the blue light was turned off, the MFE persisted for ~35 s, and then gradually faded as the mScarlet3 fluorescence recovered.

These experiments taught us several important facts. First, they established that a relatively long-lived FMN photoproduct was necessary for the MFE. Second, the magnet did not act on the blue-mediated FMN photochemistry, but rather on the interaction of the FMN photoproduct with yellow-excited mScarlet3. Third, the MFE did not require blue light interaction with mScarlet3 itself (or with mScarlet3 photoproducts). We estimated the diffusion coefficient of the FMN photoproduct from $D \approx x^2/2t$, where x = 200 μm was the distance between the blue laser spot and the mScarlet3-labeled bead, and t = 35 s was the delay between blue illumination and onset of the MFE. This estimate gave $D \approx 570$ $\mu m^2/s$, broadly consistent with diffusion of a small molecule in water (due to the complex geometry, this estimate for D is likely only accurate within a factor of ~2).

mScarlet3 MFE Requires Both Oxidized and Reduced FMN. Optically excited FMN is a potent oxidizing agent, readily forming the semiquinone radical. In the absence of an exogenous electron donor, the ribityl chain of another FMN molecule donates an electron.[12,13] The semiquinone radical spontaneously disproportionates to re-form FMN and the doubly reduced $FMNH_2$ (Figure 4A).[13−17] We speculated that photochemically produced $FMNH_2$ might mediate the MFE.

To test this hypothesis, we produced $FMNH_2$ by illuminating FMN (3.3 mM) with high intensity 488 nm light (60 W/cm^2), in a solution also containing mScarlet3 (100 μM). As before, we used yellow light (561 nm, 10 W/cm^2) to probe the mScarlet3 fluorescence. To our surprise, the mScarlet3 fluorescence rapidly quenched to almost zero, and the MFE was initially not detectable (Figure 4B). This observation seemed to contradict our hypothesis. However, after the blue light was turned off, the fluorescence gradually increased, and the MFE appeared, growing proportionally with the fluorescence. We hypothesized that the FMN concentration increased during this interval via a combination of reoxidation of $FMNH_2$ by ambient oxygen and diffusion of FMN from the nonilluminated regions of the sample into the observation region; and that perhaps it was necessary to have both $FMNH_2$ and FMN in the solution to observe the MFE.

To test this revised hypothesis, we produced $FMNH_2$ by illuminating FMN with 488 nm light in the presence of 5 mM ethylenediaminetetraacetic acid (EDTA), an efficient electron donor for photoreduction of FMN.[13,16,18] We performed the photoreduction in an anaerobic glovebox to avoid reoxidation. Absorption spectra confirmed photoreduction of FMN to $FMNH_2$ (Figure 4C). We then prepared mixtures with degassed solutions containing well-defined concentrations of $FMNH_2$, FMN, and mScarlet3 under anaerobic conditions (Figure 4D), and sealed the samples for characterization.

We used a home-built apparatus to interleave measurements of the absorption and fluorescence emission spectra while exciting the mScarlet3 with yellow light and varying the magnetic field (Figure 4E). In a sample of 30 μM $FMNH_2$ and 6 μM mScarlet3 (and no FMN), illumination with 561 nm light (1.7 W/cm^2) caused a rapid quenching of the mScarlet3

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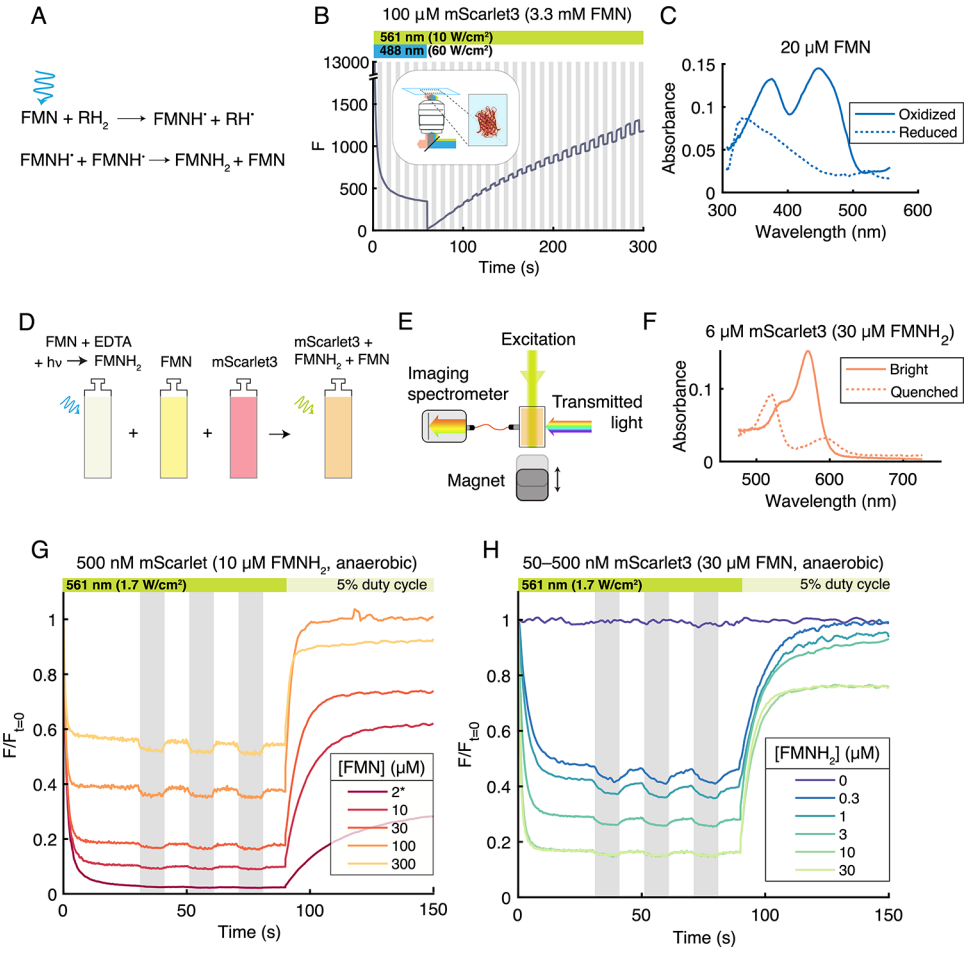


Figure 4. The mScarlet3 MFE requires both oxidized and reduced FMN. (A) Photoexcitation of FMN produces $FMNH_2$. (B) Fluorescence of purified mScarlet3 with an excess of FMN and high-intensity blue light. The fluorescence quenched almost completely (residual fluorescence during blue illumination is the tail of the FMN emission and autofluorescence due to the high blue intensity). Upon turning off the blue light, the mScarlet3 fluorescence recovered and the MFE increased proportionally. (C) Absorption spectra of FMN in its oxidized and photoreduced ($FMNH_2$) states. (D) Samples of $FMNH_2$, FMN, and mScarlet3 were prepared separately in anaerobic conditions, and then combined and measured under 561 nm illumination. (E) Top view of the experimental apparatus. Magnetic field-dependent fluorescence and absorption spectra were recorded on a spectrometer. (F) Absorption spectra of mScarlet3 and the fully photoreduced mScarlet3 product (presumably $mScarlet3^{\cdot-}$ radical anion) in the presence of $FMNH_2$. (G, H) mScarlet3 fluorescence quenching, MFE, and recovery in samples with indicated concentrations of FMN and $FMNH_2$. All fluorescence traces are normalized to their initial value. A 5% duty cycle of yellow light (25 ms every 0.5 s) was used to image the fluorescence recovery. Fluorescence during this epoch was scaled 20× to account for the lower illumination intensity. Asterisk indicates FMN concentration determined from absorption measurements (Experimental Section; Figure S3).

fluorescence to almost zero. The absorption spectrum of the mScarlet3 photoproduct was distinct from unquenched mScarlet3 (Figure 4F) with both a blue-shifted and red-shifted peak. We ascribe this spectrum to a photoreduction of mScarlet3 by $FMNH_2$, most likely to form the $mScarlet3^{\cdot-}$ radical anion. The putative $mScarlet3^{\cdot-}$ radical anion was not fluorescent under 561 nm excitation.

We then systematically varied the concentrations of FMN and $FMNH_2$ in mScarlet3 solutions and recorded the magnetic field-dependent fluorescence (Figure 4G,H). When [$FMNH_2$] was 1 μM or less, we used 50 nM mScarlet3 to ensure that $FMNH_2$ was in excess; otherwise, we used 500 nM mScarlet3. Complete photoreduction from FMN to $FMNH_2$ is difficult,[19] so for the samples with low [FMN] we determined the concentrations of each flavin species from absorption spectra (Figure S3). In contrast to previous experiments, the MFE only required yellow light.

Increasing [$FMNH_2$] decreased the steady-state mScarlet3 fluorescence, and increasing [FMN] counteracted this effect. The MFE was largest in absolute magnitude when these two effects were balanced so the steady-state fluorescence was approximately half its initial value. At fixed [$FMNH_2$], increasing [FMN] increased the rate of initial quenching and also the rate of magnetic field-induced switching (Figure 4G). In the absence of $FMNH_2$, FMN had no effect on mScarlet3 and did not induce an MFE (Figure 4H, purple trace), as we observed previously (e.g., Figure 2B).

After 90 s of yellow illumination, we decreased the yellow intensity to 5% (85 mW/cm^2), to monitor the recovery of mScarlet3 fluorescence in close-to-dark conditions. The rate of mScarlet3 recovery in dim yellow light depended on [FMN] and was largely independent of [$FMNH_2$] (except for the small effect of residual yellow-light + $FMNH_2$-mediated quenching).

In anaerobic conditions, the MFE persisted overnight (Figure S4A), establishing that the active FMN photoproduct was stable in the absence of oxygen, while stirring the solution in air immediately abolished the MFE. All evidence was consistent with $FMNH_2$ as the photoproduct responsible for

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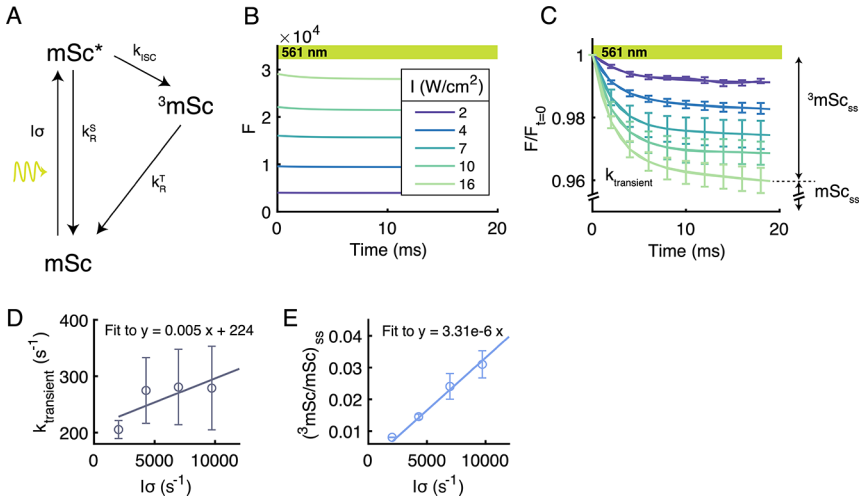


Figure 5. Illumination of mScarlet3 populates a transient dark state. (A) Three-state model of intersystem crossing (ISC) in mScarlet3. (B) Onset of high-intensity illumination of mScarlet3 immobilized on an agarose Ni-NTA bead induced a rapid and reversible slight decrease in fluorescence. (C) Data from (B), normalized to initial fluorescence, with exponential + linear fits. More intense illumination led to a larger fractional decrease in fluorescence. The relative amplitudes of the fluorescence dip and the fluorescence steady-state correspond to the relative triplet and ground-state populations (assuming negligible singlet excited state population). Traces are averaged over 18 flashes for each of $n = 3$ beads. Fitting the (D) rate constant and (E) amplitude of the fluorescence dip implied $k_R^T \approx 224\ s^{-1}$ and $k_{ISC} \approx 1.9 \times 10^5\ s^{-1}$. Error bars denote s.e.m.

photoinduced mScarlet3 quenching and the MFE. To rule out the possibility of species other than $FMNH_2$ and FMN contributing to the MFE, we tested mixtures of mScarlet3 with EDTA alone (Figure S4B), as well as with FMN photoproducts lumichrome and H_2O_2 (Figure S4C).[13,17,18] None of these mixtures showed an MFE. We measured the MFE at several pH values to determine whether the reductant was $FMNH_2$ or $FMNH^-$, as $FMNH_2$ has a pK_a of 6.7.[19] The mScarlet3 fluorescence quenched more at lower pH, suggesting that $FMNH_2$ was the relevant species (Figure S5).

Together, these results imply that $FMNH_2$ electrochemically reduced the optically excited mScarlet3 to form a non-fluorescent protein photoproduct, and FMN reoxidized this photoproduct to reform ground-state mScarlet3. In this model, electron transfer from $FMNH_2$ to mScarlet3 likely proceeded through a radical-pair intermediate, whose branching between recombination and separation was magnetic field sensitive.

mScarlet3 Photoreduction Likely Proceeds through an mScarlet3 Triplet State. Two observations led us to suspect that $FMNH_2$ electrochemically reduced photogenerated mScarlet3 triplet (^{3}mSc) rather than the excited mScarlet3 singlet (mSc*). First, under our illumination conditions, the population of mSc* was very low. The mSc* lifetime is 4 ns,[11] and at an excitation intensity of 1.7 W/cm^2 at 561 nm, the per-molecule excitation rate was $I\sigma = 1652\ s^{-1}$ (ε_{561} = 90,480 $M^{-1}\ cm^{-1}$). Thus, the excited-state to ground-state ratio mSc*/mSc $\approx 6 \times 10^{-6}$. At 10 μM $FMNH_2$, and a measured rate of initial quenching of 0.98 s^{-1}, the implied bimolecular rate constant if the reaction proceeded from mSc* would be $8.3 \times 10^{10}\ M^{-1}\ s^{-1}$, substantially greater than the diffusion-limited aqueous bimolecular rate constant, $6.6 \times 10^9\ M^{-1}\ s^{-1}$.[20] Hence a reaction from a long-lived triplet which could accumulate to higher concentration was more likely.

Second, we noted that the magnet decreased the mScarlet3 fluorescence. A magnetic field prolongs the lifetime of the initial spin state of a SCRP. A singlet SCRP is more likely to recombine, releasing ground-state mScarlet3, which can fluoresce, while a triplet SCRP is more likely to diffusively separate, building up a population of non-fluorescent $mSc^{\cdot-}$ radical anion. Thus, we surmised that the SCRP was likely born in a triplet state. The electron-transfer reaction usually conserves spin, so one of the reactants was likely a triplet. $FMNH_2$ has a singlet ground state, so the mScarlet3 was likely in a triplet state.

We measured the mSc* → ^{3}mSc intersystem crossing and ^{3}mSc → mSc recovery rates for mScarlet3 (with no flavin or other redox factors present), which we modeled as a three-state system (Figure 5A). In an anaerobic sample of mScarlet3 bound to beads (to eliminate effects of diffusion), we applied pulses of 561 nm light at different intensities (Figure 5B) and recorded the millisecond-time scale fluorescence dynamics (Figure 5C). We observed a reversible intensity-dependent initial dip in the fluorescence, which we attributed to populating the mScarlet3 triplet state. We quantified the rate (Figure 5D) and amplitude (Figure 5E) for this initial dip as a function of the mScarlet3 excitation rate ($I\sigma$), where I is the excitation intensity (photons/cm^2/s) and σ is the absorption cross section at 561 nm (cm^2). From these, we determined $k_R^T \approx 224\ s^{-1}$ and $k_{ISC} \approx 1.9 \times 10^5\ s^{-1}$ (Supporting Information).

Quantitative Model of the mScarlet3/$FMNH_2$/FMN Magnetic Field Effect. Combining all the above results, we developed a model of the mScarlet3 and FMN magnetic field effect (Figure 6A). In a solution initially containing only FMN, the blue illumination produced $FMNH_2$. Alternatively, the $FMNH_2$ can be produced separately and added to the solution under anaerobic conditions. Yellow excitation populated the mScarlet3 triplet state (^{3}mSc, Figure 5). In the key reaction step, $FMNH_2$ donated an electron to the mScarlet3 triplet state, producing an initially triplet spin-correlated radical pair, $^3[mSc^{\cdot-} + FMNH^{\cdot}]$. The magnetic field suppressed intersystem crossing to the singlet. The radical pair either recombined (more likely if in a singlet) or separated (more likely if in a triplet). Once separated, FMN reoxidized $mSc^{\cdot-}$ back to mScarlet3. This model contrasts with the MFE in interactions of FMN with a non-fluorescent protein backbone, where the optically excited FMN is proposed to accept an electron from the protein to produce a SCRP.[7]

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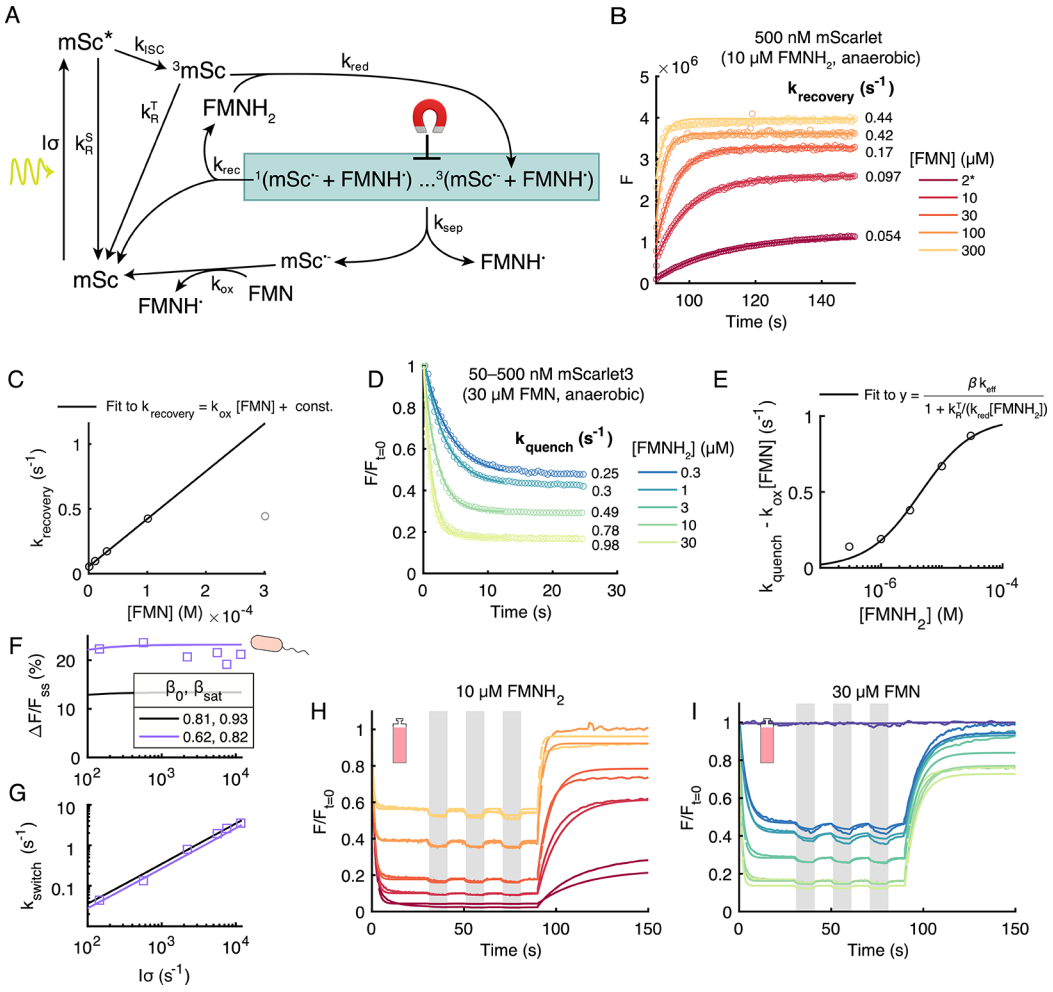


Figure 6. Quantitative model of giant MFE in mixture of mScarlet3, FMN, and $FMNH_2$. (A) Proposed mechanism. A yellow photon excites mScarlet3. The excited singlet state either fluoresces or undergoes ISC to produce the mScarlet3 triplet. Blue light drives photoreduction of FMN to $FMNH_2$ (reactions shown in Figure 4A). $FMNH_2$ donates an electron to triplet mScarlet3 to form a triplet-born spin-correlated radical pair (SCRP). The magnetic field controls the rate of ISC in the SCRP. Singlet products preferentially recombine to reform ground-state mScarlet3. Triplet products diffuse apart to release a non-fluorescent mScarlet3$^{\cdot-}$ radical anion. FMN reoxidizes the mScarlet3$^{\cdot-}$ back to the mScarlet3 ground state. (B) Exponential + constant fits of the fluorescence recovery for the FMN titration in Figure 4G. (C) Linear fit of $k_{recovery}$ as a function of [FMN] determines the rate constant k_{ox} of mScarlet3$^{\cdot-}$ reoxidation by FMN. (D) Exponential + constant fits of the fluorescence quenching for the $FMNH_2$ titration in Figure 4H. (E) Fitting k_{quench} as a function of [$FMNH_2$] determines parameters β_0 and k_{red} (Supporting Information). (F) Comparison of measured and predicted MFE amplitude as a function of light intensity in *E. coli*. (G) Comparison of measured and predicted MFE upstroke switching rate as a function of light intensity in *E. coli*. (H, I) Comparison of measured and predicted fluorescence dynamics for the experiments in Figures 4G,H. All rate constants in (F–I) were identical. The magnetic field-dependent SCRP separation fractions β_0 and β_{sat} were fit separately for *E. coli* and in vitro data.

We wrote the rate equations for this model and determined the rate constants from our data. The key steps are outlined below, the details are in the Supporting Information. Our measurements did not directly probe the SCRP dynamics, so we captured the MFE via the magnetic field-dependent separation fraction β(B) and recombination fraction, 1-β(B), from the SCRP states. We used β as a fitting parameter.

We first fit the rates of fluorescence recovery under dim yellow light, $k_{recovery}$, in the FMN titration data (Figure 6B). From this, we calculated the bimolecular rate constant, k_{ox}, of mSc$^{\cdot-}$ reoxidation by FMN (Figure 6C). We then fit the rates of initial fluorescence quenching, k_{quench}, in the $FMNH_2$ titration data (Figure 6D). By combining k_{ISC}, determined above, with the dependence of k_{quench} on [$FMNH_2$], we determined (1) the bimolecular rate constant k_{red}, for ^{3}mSc reduction by $FMNH_2$ (Figure 6E), and (2) the SCRP separation fraction at zero magnetic field, $\beta_0 \equiv \beta(B = 0)$. We determined $\beta_{sat} \equiv \beta(B_{sat})$ by fitting the numerical model to our data on $\Delta F/F_{ss}$ and $\Delta F/F_{t=0}$ as a function of [FMN] and [$FMNH_2$] (i.e., Figure 4G,H). The best-fit values were β_0 = 0.81 and β_{sat}= 0.93. The fit parameters are given in Table 1 and the model is available as Supplementary Code.

Our model explained many features of our data, both in *E. coli* and with purified protein. In *E. coli*, upon onset of illumination at $I\sigma$ = 11,400 s^{-1}, the mScarlet3 fluorescence initially dropped by more than 100-fold, from 13,000 to 80 counts. Our model reproduced the near-independence of MFE amplitude ($\Delta F/F_{ss}$) on illumination intensity under conditions of strong quenching (Figure 6F, compare to Figure 1E), the linear dependence of magnetic field-induced switching rate (k_{switch}) on illumination intensity (Figure 6G, compare to Figure 1G), and the independence of k_{switch} on magnetic field strength (acting through $\beta(B)$, Figure S6, compare to Figure 1H).

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Table 1. Fit Parameters for the Kinetic Model of the Photocycle

parameter	value	description	source
σ	3.5×10^{-16} cm^2	mScarlet3 absorption cross section at 561 nm	ref 11
k_R^S	2.5×10^8 s^{-1}	inverse of the mScarlet3 excited-state lifetime	ref 11
k_{ISC}	1.85×10^5 s^{-1}	mScarlet3 intersystem crossing rate	fit from Figure 5E
k_R^T	224 s^{-1}	inverse of the mScarlet3 triplet lifetime	fit from Figure 5D
k_{ox}	3375 M^{-1} s^{-1}	oxidation of mScarlet3 radical anion by FMN	fit from Figure 6C
β_0	0.81		fit to numerical model
β_{sat}	0.93	*in vitro*SCRP separation fraction at magnetic field $B = B_{sat}$	fit to numerical model
k_{red}	5×10^7 M^{-1} s^{-1}	reduction of triplet mScarlet3 by $FMNH_2$	fit from Figure 6E
k_{dis}	5×10^9 M^{-1} s^{-1}	rate constant for $FMNH^{\cdot}$ disproportionation	ref 15

Using the parameters determined from our experiments in purified proteins, the maximal predicted MFE in vitro was 14% across all illumination intensities and concentrations of FMN and $FMNH_2$. To match the larger MFE (21%) observed in *E. coli*, we decreased the values of β to $\beta_0 = 0.62$ and $\beta_{sat} = 0.82$. These decreases in β can be attributed to a greater bias toward recombination vs separation in *E. coli* compared to in vitro (Supporting Information). This modification is plausible, considering the influence of nanoscale confinement and crowding in the *E. coli* cytoplasm.

For the experiments with purified protein, the model reproduced the dependence of quenching amplitude ($F_{ss}/F_{t=0}$) on FMN and $FMNH_2$ concentrations (Figure S7). We then simulated the complete experiment of Figure 4F,G, including the initial quenching, the magnetic field-dependent changes in fluorescence, and the recovery of fluorescence under dim yellow light. By varying the nominal flavin concentrations by <30% (to account for possible background redox processes) we achieved close overlay of the predicted and measured traces (Figure 6H,I).

Implications of the Model. We used the kinetic model to explore how to maximize the MFE magnetic field sensitivity and kinetics. The most easily controlled parameters are [FMN], [$FMNH_2$], and light intensity ($I\sigma$). Surprisingly, we found that when the MFE was small, variation in all three parameters caused the absolute MFE, $\Delta F/F_{t=0}$, to vary along a single universal curve parametrized by $x \equiv F_{ss}/F_{t=0}$ i.e., the steady-state residual fluorescence. Specifically, in the limit of small $\Delta\beta \equiv \beta(B) - \beta(0)$, the kinetic equations predicted:

$$\Delta F/F_{t=0} \approx \frac{\Delta\beta}{\beta_0} x(1-x) \quad (1)$$

$$\Delta F/F_{ss} \approx \frac{\Delta\beta}{\beta_0} (1-x) \quad (2)$$

(here we have preserved the sign convention $\Delta F \equiv F(0) - F(B)$). The derivation is in the Supporting Information. Eq 1 confirms our earlier observation that the absolute MFE was maximized around $x = 1/2$. In vitro data with variable [FMN], [$FMNH_2$] (Figure 4G,H) and $I\sigma$ (Figure S8) all fell close to eq 1, confirming its universality (Figure 7A). Eq 2 predicts that the relative MFE is largest when the fluorescence is most strongly quenched ($x \approx 0$). In our experiments in *E. coli*, the steady-state residual fluorescence ranged from $x = 2.6\%$ at $I = 0.3$ to $x = 0.6\%$ at $I = 21$ W/cm^2, explaining why we did not detect the intensity-dependent change in $\Delta F/F_{ss}$. This analysis presumes that there is no fluorescent background. Presence of a small fluorescent background shifts the maximal value of $\Delta F/F_{ss}$ to higher x, while having little effect on the maximal value of $\Delta F/F_{t=0}$.

The different conditions for maximizing absolute vs relative MFE highlight a general fact: different chemical quantities may have their magnetic field sensitivity maximized under different conditions. We used the model to predict the time-courses of both observed and unobserved quantities (Figure S9). Figures 7B–F show the dependence on [FMN] and [$FMNH_2$] of $F_{ss}/F_{t=0}$, $\Delta F/F_{ss}$, $\Delta F/F_{t=0}$, the fractional change in mScarlet3$^{\cdot-}$ ($-\Delta mSc^{\cdot-}/mSc^{\cdot-}_{ss}$), and the magnetic field-induced switching rate, k_{switch}. Figures S10 and S11 show the dependence of the same variables on the excitation rate, $I\sigma$, and either [FMN] or [$FMNH_2$].

One may wish to use the MFE to detect time-dependent magnetic fields. The switching rate, k_{switch}, is the sum of the rates into and out of the $mSc^{\cdot-}$ state. Under continuous low-intensity illumination, k_{switch} depends linearly on $I\sigma$ and [FMN]. Increasing [$FMNH_2$] increases k_{switch} up until [$FMNH_2$] ≈ 5.5 μM. At high light intensity ($I > 10^6$ W/cm^2), the MFE switching follows a multiexponential, with components corresponding to k_{ISC} and $k_{red} \times$ [$FMNH_2$]. The highest switching rate in our simulations was 1.16×10^5 s^{-1} (Figure S12), though our model might fail at such high light intensities, where other photoreactions could come into play.

An alternate approach to measuring fast (and periodic) magnetic field dynamics is to pulse the illumination in synchrony with the oscillations in magnetic field, and to measure the rate of accumulation of $mSc^{\cdot-}$, i.e. as a light-gated integrator. In this case, one should set [FMN] = 0 so that the $mSc^{\cdot-}$ is long-lived. The time-resolution is set simply by the rate of reaction between ^{3}mSc and $FMNH_2$ to form the SCRP, presuming that the subsequent dynamics are fast. At close-to-saturating [$FMNH_2$] = 1.4 mM,[21,22] this rate is predicted to be 7×10^4 s^{-1}, implying the ability to use this "lock-in" scheme to detect magnetic fields at up to tens of kHz. The maximum concentration of $FMNH_2$ might be increased via addition of hydrotropic agents such as nicotinamide,[23,24] though it is not known how well our model will work at concentrations well outside the range used to determine its parameters.

Modeling the SCRP Dynamics. Our kinetic model places no constraints on the SCRP separation fraction, $\beta(B)$, but in reality this function is governed by the SCRP spin dynamics and reaction kinetics. The SCRP dynamics can be described at varying levels of theory. The Haberkorn master equation provides a fully quantum description,[25] while the rate-based description of Hayashi and Nagakura provides a useful approximation.[26] The model of Hayashi and Nagakura captures the fact that in strong magnetic field, the $T_\pm$ sublevels and the T_0 sublevel are not chemically equivalent: $T_\pm$ are energetically forbidden from interconverting with the singlet state, S, while T_0 and S continue to interconvert in the presence of a strong field.[26] This model predicts that in the limit of fast hyperfine-mediated intersystem crossing, $\beta_0 = 1/\left(1 + \frac{\eta}{2}\right)$, where $\eta \equiv k_{rec}/k_{sep}$ is the ratio of

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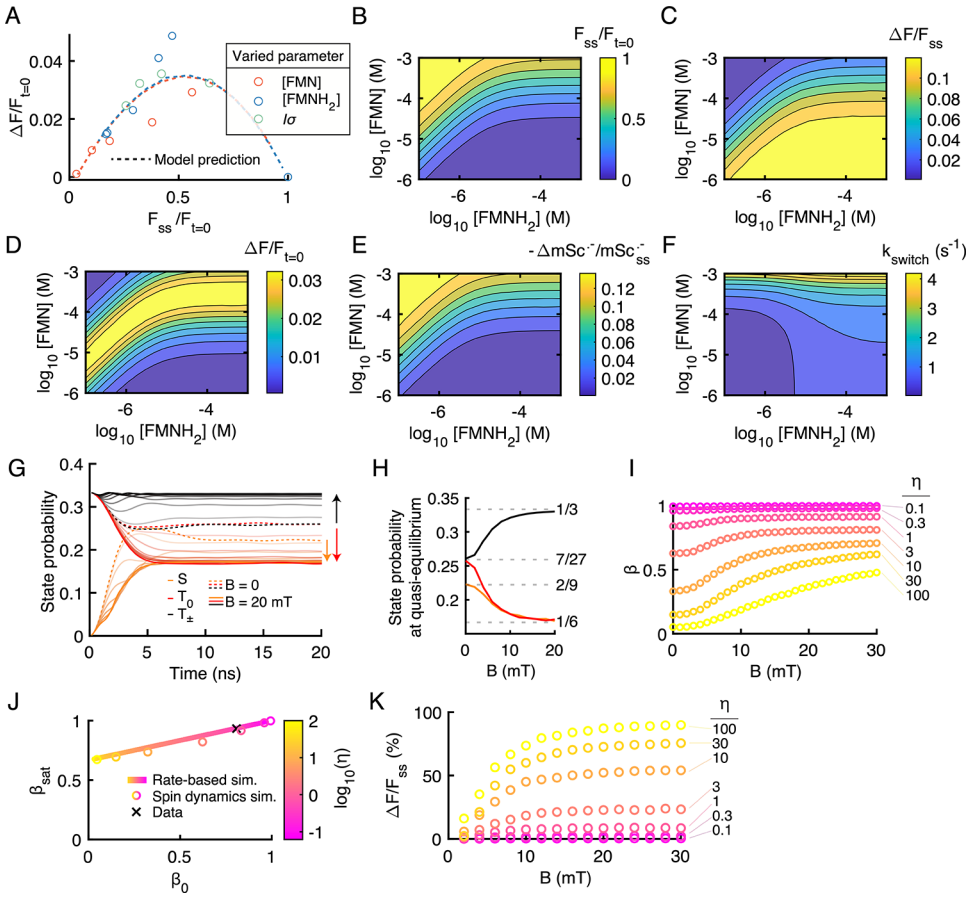


Figure 7. Model predictions for the MFE. (A) Universal curve predicts relation between absolute MFE, $\Delta F/F_{t=0}$, and residual steady-state fluorescence, $x = F_{ss}/F_{t=0}$, when the MFE is small. Variation in [FMN], [$FMNH_2$] and light intensity all led to variations in MFE along this curve. (B–F) Predicted effects of [$FMNH_2$] and [FMN] on (B) $F_{ss}/F_{t=0}$, (C) $\Delta F/F_{ss}$, (D) $\Delta F/F_{t=0}$, (E) $(-\Delta mSc^{\cdot-}/mSc^{\cdot-}_{ss})$, and (F) k_{switch} for the MFE upstroke. Illumination intensity was assumed to be $I\sigma = 1652\ s^{-1}$, corresponding to 1.7 W/cm^2 at 561 nm. (G) Simulated ensemble-averaged spin-state probabilities for SCRPs initially equally populated in the T_0, T_+, and T_- states. The hyperfine field was Gaussian distributed in three dimensions, with standard deviation 2 mT for each electron spin, and the external magnetic field was varied from 0 − 20 mT. Arrows point from low to high external magnetic field. (H) Magnetic field-dependent spin state probabilities at steady state (20 ns on panel (G)). Probabilities for zero and saturating magnetic fields agree with analytical results. (I) Simulated SCRP separation fraction, β, as a function of magnetic field strength, at varying values of $\eta = k_{rec}/k_{sep}$. (J) Relation between values of β under saturating-field vs zero-field, for different values of η. The rate-based simulation (solid line) has intercept 2/3 and slope 1/3. Semiclassical spin dynamics simulation (circles) and experimental result (x) lie near this line. (K) Predicted MFE amplitude as a function of magnetic field for varying values of η. In these simulations, we kept constant [FMN] = 0.38 μM, [$FMNH_2$] = 6 μM.

recombination and separation rate constants. The model further predicts that $\beta_{sat} = 2/3 + 1/3\beta_0$ (Figure 7J). This simple result is a strong constraint on the maximal MFE amplitude. Remarkably, the values of β we determined by fitting the kinetic photocycle model to our in vitro data, $\beta_0 = 0.81$ and $\beta_{sat} = 0.93$, lie within 1% of this line (i.e., $2/3 + 1/3 \times 0.81 = 0.94$), establishing good concordance of model and experiment.

The Hayashi and Nagakura model has two limitations: (1) as a rate-based model it may miss effects due to the coherent electron spin dynamics; and (2) it only predicts the limiting dynamics in zero and saturating magnetic fields, not the intermediate cases. To address these limitations, we calculated $\beta(B)$ by simulating the quantum dynamics and reaction kinetics of the SCRP spins (Supporting Information). Briefly, SCRP spins were initialized with equal populations in T_0, T_+, and T_- states, and set in motion precessing around local random hyperfine fields plus the external magnetic field. The singlet and triplet probabilities were calculated as a function of time. In the absence of chemical reactions, the ensemble-average state occupancies agreed with prior analytical predictions (Figure 7G,H).[27,28] To introduce chemical reactions, at each time-step we let each SCRP either separate, with a rate constant k_{sep} (assumed to be the same for singlet and triplet states), or recombine, with a rate constant $k_{rec}(1 - \alpha(t))$, where $(1 - \alpha(t))$ is the time-dependent probability of being in the singlet state (Figure S13).

We tallied the fraction of SCRPs that separated to determine $\beta(B)$. We then repeated this process for a range of values of $\eta \equiv k_{rec}/k_{sep}$ (Figure 7I). These semiclassical spin dynamics simulations gave values of β_{sat} and β_0 which were very close to the rate-based result $\beta_{sat} \approx 2/3 + 1/3\beta_0$ (Figure 7J). We then input the simulated $\beta(B)$ values into our model of the photocycle (Figure 6A) to determine the $\Delta F/F_{ss}$ vs B curves for different values of η (Figure 7K). As k_{sep} approached zero (e.g., large η), $\Delta F/F_{ss}$ approached 100%.

DISCUSSION

We determined the mechanism by which small magnetic fields can modulate the fluorescence of mScarlet3. The reaction

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requires the simultaneous presence of fully reduced $FMNH_2$ and oxidized FMN, and involves a photocycle comprising reduction of triplet-state mScarlet3 by $FMNH_2$, and reoxidation of $mScarlet3^{\bullet-}$ by FMN. A numerical model quantitatively reproduced the time-dependent fluorescence trajectories under varied conditions of FMN, $FMNH_2$, illumination intensity, and magnetic field, both in vitro and in live *E. coli*. We infer that live *E. coli* cytoplasm contains sufficient FMN and $FMNH_2$ (or perhaps other oxidant and reductant), so the MFE does not require blue light. Previous work has shown that other redox cofactors such as cinnavalininate and WST-8 can also enable the MFE.[1]

Some molecular details in our model remain uncertain. The molecular structure of the putative $mScarlet3^{\bullet-}$ is not known. Many red fluorescent proteins—KillerRed, mRFP, DsRed, and mKate2—accept electrons upon photoexcitation,[29−31] suggesting the presence of a common mechanism. Electron spin resonance (ESR) studies may provide insights into the nature of the radical anion. The detailed mechanism underlying electron-transfer between $FMNH_2$ and ^{3}mSc is also unknown. Fluorescent proteins undergo multiple light-driven redox reactions, some involving direct electron-transfer between the chromophore and an external redox species, and some involving hopping mediated by other amino acids in the protein.[31] Further mutagenesis, transient absorption, and computational studies will be needed to elucidate the details of this reaction step.

In organic donor−acceptor complexes, the MFE can be enhanced by tethering the groups together,[5,32] or by confinement in micelles.[10,33] In the context of our model, these effects can be understood as decreasing the rate of SCRP separation (i.e., increasing η), which would decrease β_0 and β_{sat} and would increase $\beta_{sat} - \beta_0$. Thus, tethering the flavin cofactor or nanoconfinement (in micelles, vesicles, or pores in a gel) may substantially enhance the MFE amplitude. Tethering a fluorescent protein to a flavin-binding domain may enable development of magneto-switchable proteins in mammalian cells, a necessary step for opaque-tissue applications of magnetogenetics.

To increase the magnetic field sensitivity, one could decrease the strength of the hyperfine coupling at the locations of the spins. It may be possible to identify redox cofactors that have weaker hyperfine couplings than $FMNH^{\bullet}$, e.g., by replacing the H with deuterium at position N5 in FMN.[34] Whether directed evolution can modulate the hyperfine environment for the electron spin on $mSc^{\bullet-}$ remains an open question.

CONCLUSIONS

A detailed mechanistic understanding of spin states in fluorescent proteins may lead to new microscopy and sensing techniques.[35,36] For example, a recent study demonstrated fluorescence-detected electron-spin resonance via optical pumping of the triplet state in yellow fluorescent protein (YFP).[37] Other recent studies showed fluorescence-detected electron-spin resonance in a modified LOV domain, magLOV,[38] in mScarlet in a live nematode,[39] and in a cryptochrome.[40] In organic solutions showing magnetic-sensitive fluorescence, imaging through a scattering medium was achieved by sweeping a magnetic null through the sample.[41] Development of protein-based spintronics requires a detailed understanding of both the quantum spin dynamics and the kinetics of the photocycles in which these spin dynamics are embedded.

Photoredox reactions are common in proteins: cryptochromes and photolyases contain flavin cofactors,[42] and protein backbones can donate electrons to photoexcited FMN.[7,42] Among fluorescent proteins, GFP can donate an electron from the excited state, undergoing oxidative redding.[43,44] Under anaerobic conditions, GFP may also accept an electron.[44−46] We suspect that many proteins with these properties may be magnetic field-sensitive or could be engineered to show magnetosensitivity under suitable conditions.

It remains controversial whether radical pair-based magnetic field effects have any biological function.[47,48] The half-maximal field of 5.5 mT in our experiments is approximately 100-fold larger than Earth's magnetic field, so our experiments do not directly address this controversy. Regardless, the existence of strong MFEs in proteins under physiological conditions suggests that such effects may be more widespread than currently appreciated. An exciting future opportunity is to use the power of protein design and directed evolution to tune the MFE amplitude, magnetic field sensitivity, and switching kinetics; and to couple the MFE to downstream biochemical cascades.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c03997.

Experimental details, materials, and methods; characterization of the MFE, control experiments, and additional simulation predictions; mathematical modeling and fitting of the MFE and spin dynamics simulations; and code for the photocycle kinetic model and SCRP spin dynamics simulations is available at https://github.com/adamcohenlab/xiang2025mfe (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Hayward, R. F.; Lazzari-Dean, J. R.; York, A. G.; Ingaramo, M.Magnetic Control of GFP-like Fluorescent Proteins. In *Proc. SPIE PC12863, Quantum Effects and Measurement Techniques in Biology and Biophotonics*; SPIE, 2023.

(2) Turro, N. J.; Kraeutler, B. Magnetic Field and Magnetic Isotope Effects in Organic Photochemical Reactions. A Novel Probe of Reaction Mechanisms and a Method for Enrichment of Magnetic Isotopes. *Acc. Chem. Res.* **1980**, *13* (10), 369−377.

(3) Turro, N. J.; Kraeutler, B. Magnetic Isotope and Magnetic Field Effects on Chemical Reactions. Sunlight and Soap for the Efficient Separation of Carbon-13 and Carbon-12. *J. Am. Chem. Soc.* **1978**, *100* (23), 7432−7434.

(4) Cao, H.; Fujiwara, Y.; Haino, T.; Fukazawa, Y.; Tung, C. H.; Tanimoto, Y. Magnetic Field Effects on Intramolecular Exciplex Fluorescence of Chain-Linked Phenanthrene and N, N-Dimethylaniline: Influence of Chain Length, Solvent, and Temperature. *Bull. Chem. Soc. Jpn.* **1996**, *69* (10), 2801−2813.

(5) Lee, H.; Yang, N.; Cohen, A. E. Mapping Nanomagnetic Fields Using a Radical Pair Reaction. *Nano Lett.* **2011**, *11*, 5367−5372.

(6) Ikeya, N.; Woodward, J. R. Cellular Autofluorescence Is Magnetic Field Sensitive. *Proc. Natl. Acad. Sci. U.S.A.* **2021**, *118* (3), No. e2018043118.

(7) Kattnig, D. R.; Evans, E. W.; Déjean, V.; Dodson, C. A.; Wallace, M. I.; Mackenzie, S. R.; Timmel, C. R.; Hore, P. J. Chemical Amplification of Magnetic Field Effects Relevant to Avian Magnetoreception. *Nature Chem.* **2016**, *8* (4), 384−391.

(8) Xu, J.; Jarocha, L. E.; Zollitsch, T.; Konowalczyk, M.; Henbest, K. B.; Richert, S.; Golesworthy, M. J.; Schmidt, J.; Déjean, V.; Sowood, D. J. C.; Bassetto, M.; Luo, J.; Walton, J. R.; Fleming, J.; Wei, Y.; Pitcher, T. L.; Moise, G.; Herrmann, M.; Yin, H.; Wu, H.; Bartölke, R.; Käsehagen, S. J.; Horst, S.; Dautaj, G.; Murton, P. D. F.; Gehrckens, A. S.; Chelliah, Y.; Takahashi, J. S.; Koch, K.-W.; Weber, S.; Solov'yov, I. A.; Xie, C.; Mackenzie, S. R.; Timmel, C. R.; Mouritsen, H.; Hore, P. J. Magnetic Sensitivity of Cryptochrome 4 from a Migratory Songbird. *Nature* **2021**, *594* (7864), 535−540.

(9) Takiff, L.; Boxer, S. G. Phosphorescence from the Primary Electron Donor in Rhodobacter Sphaeroides and Rhodopseudomonas Viridis Reaction Centers. *Biochimica et Biophysica Acta (BBA) - Bioenergetics* **1988**, *932*, 325−334.

(10) Steiner, U. E.; Ulrich, T. Magnetic Field Effects in Chemical Kinetics and Related Phenomena. *Chem. Rev.* **1989**, *89* (1), 51−147.

(11) Gadella, T. W. J.; Van Weeren, L.; Stouthamer, J.; Hink, M. A.; Wolters, A. H. G.; Giepmans, B. N. G.; Aumonier, S.; Dupuy, J.; Royant, A. mScarlet3: A Brilliant and Fast-Maturing Red Fluorescent Protein. *Nat. Methods* **2023**, *20* (4), 541−545.

(12) Holmström, B.; Oster, G. Riboflavin as an Electron Donor in Photochemical Reactions. *J. Am. Chem. Soc.* **1961**, *83* (8), 1867−1871.

(13) Song, S.-H.; Dick, B.; Penzkofer, A. Photo-Induced Reduction of Flavin Mononucleotide in Aqueous Solutions. *Chem. Phys.* **2007**, *332* (1), 55−65.

(14) Penzer, G. R. The Chemistry of Flavins and Flavoproteins: Aerobic Photochemistry. *Biochem. J.* **1970**, *116* (4), 733−743.

(15) Holmström, B. Flash Photoreduction of Flavin Mono Nucleotide in Neutral Solutions: Absolute Absorptivity of the Semiquinone. *Photochem. & Photobiology* **1964**, *3* (2), 97−114.

(16) Vaish, S. P.; Tollin, G. Flash Photolysis of Flavins. V. Oxidation and Disproportionation of Flavin Radicals. *J. Bioenerg Biomembr* **1971**, *2* (2), 61−72.

(17) Holzer, W.; Shirdel, J.; Zirak, P.; Penzkofer, A.; Hegemann, P.; Deutzmann, R.; Hochmuth, E. Photo-Induced Degradation of Some Flavins in Aqueous Solution. *Chem. Phys.* **2005**, *308* (1−2), 69−78.

(18) Heelis, P. F.; Parsons, B. J.; Phillips, G. O.; McKellar, J. F. The photoreduction of flavins by amino acids and EDTA. A continuous and flash photolysis study. *Photochem. & Photobiology* **1979**, *30* (3), 343−347.

(19) Draper, R. D.; Ingraham, L. L. A Potentiometric Study of the Flavin Semiquinone Equilibrium. *Arch. Biochem. Biophys.* **1968**, *125* (3), 802−808.

(20) Berg, H. C. *Random Walks in Biology*; Princeton University Press: Princeton, NJ, 1993.

(21) Guo, A. C.; Jewison, T.; Wilson, M.; Liu, Y.; Knox, C.; Djoumbou, Y.; Lo, P.; Mandal, R.; Krishnamurthy, R.; Wishart, D. S. ECMDB: The E. Coli Metabolome Database. *Nucleic Acids Res.* **2012**, *41* (D1), D625−D630.

(22) Sajed, T.; Marcu, A.; Ramirez, M.; Pon, A.; Guo, A. C.; Knox, C.; Wilson, M.; Grant, J. R.; Djoumbou, Y.; Wishart, D. S. ECMDB 2.0: A Richer Resource for Understanding the Biochemistry of *E. Coli*. *Nucleic Acids Res.* **2016**, *44* (D1), D495−D501.

(23) Coffman, R. E.; Kildsig, D. O. Effect of Nicotinamide and Urea on the Solubility of Riboflavin in Various Solvents. *J. Pharm. Sci.* **1996**, *85* (9), 951−954.

(24) Orita, A.; Verde, M. G.; Sakai, M.; Meng, Y. S. A Biomimetic Redox Flow Battery Based on Flavin Mononucleotide. *Nat. Commun.* **2016**, *7* (1), 13230.

(25) Haberkorn, R. Density Matrix Description of Spin-Selective Radical Pair Reactions. *Mol. Phys.* **1976**, *32* (5), 1491−1493.

(26) Hayashi, H.; Nagakura, S. Theoretical Study of Relaxation Mechanism in Magnetic Field Effects on Chemical Reactions. *Bull. Chem. Soc. Jpn.* **1984**, *57* (2), 322−328.

(27) Schulten, K.; Wolynes, P. G. Semiclassical Description of Electron Spin Motion in Radicals Including the Effect of Electron Hopping. *J. Chem. Phys.* **1978**, *68* (7), 3292−3297.

(28) Cohen, A. E. Nanomagnetic Control of Intersystem Crossing. *J. Phys. Chem. A* **2009**, *113* (41), 11084−11092.

(29) Vegh, R. B.; Bravaya, K. B.; Bloch, D. A.; Bommarius, A. S.; Tolbert, L. M.; Verkhovsky, M.; Krylov, A. I.; Solntsev, K. M. Chromophore Photoreduction in Red Fluorescent Proteins Is Responsible for Bleaching and Phototoxicity. *J. Phys. Chem. B* **2014**, *118* (17), 4527−4534.

(30) Protasova, E. A.; Mishin, A. S.; Lukyanov, K. A.; Maksimov, E. G.; Bogdanov, A. M. Chromophore Reduction plus Reversible Photobleaching: How the mKate2 "Photoconversion" Works. *Photochem. Photobiol. Sci.* **2021**, *20* (6), 791−803.

(31) Acharya, A.; Bogdanov, A. M.; Grigorenko, B. L.; Bravaya, K. B.; Nemukhin, A. V.; Lukyanov, K. A.; Krylov, A. I. Photoinduced Chemistry in Fluorescent Proteins: Curse or Blessing? *Chem. Rev.* **2017**, *117* (2), 758−795.

(32) Lee, H.; Brinks, D.; Cohen, A. E. Two-Photon Imaging of a Magneto-Fluorescent Indicator for 3D Optical Magnetometry. *Opt. Express, OE* **2015**, *23* (21), 28022−28030.

(33) Ulrich, T.; Steiner, U. E. Magnetic-Field-Dependent Recombination Kinetics of Geminate Radical Pairs in Reversed Micelles of Variable Size. *Chem. Phys. Lett.* **1984**, *112* (4), 365−370.

(34) Rostas, A.; Einholz, C.; Illarionov, B.; Heidinger, L.; Said, T. A.; Bauss, A.; Fischer, M.; Bacher, A.; Weber, S.; Schleicher, E. Long-Lived Hydrated FMN Radicals: EPR Characterization and Implications for Catalytic Variability in Flavoproteins. *J. Am. Chem. Soc.* **2018**, *140* (48), 16521−16527.

(35) Byrdin, M.; Duan, C.; Bourgeois, D.; Brettel, K. A Long-Lived Triplet State Is the Entrance Gateway to Oxidative Photochemistry in

https://doi.org/10.1021/jacs.5c03997
J. Am. Chem. Soc. 2025, 147, 18088−18099

Green Fluorescent Proteins. *J. Am. Chem. Soc.* **2018**, *140* (8), 2897−2905.
(36) Ludvikova, L.; Simon, E.; Deygas, M.; Panier, T.; Plamont, M.-A.; Ollion, J.; Tebo, A.; Piel, M.; Jullien, L.; Robert, L.; Le Saux, T.; Espagne, A. Near-Infrared Co-Illumination of Fluorescent Proteins Reduces Photobleaching and Phototoxicity. *Nat. Biotechnol.* **2024**, *42* (6), 872−876.
(37) Feder, J. S.; Soloway, B. S.; Verma, S.; Geng, Z. Z.; Wang, S.; Kifle, B.; Riendeau, E. G.; Tsaturyan, Y.; Weiss, L. R.; Xie, M.; Huang, J.; Esser-Kahn, A.; Gagliardi, L.; Awschalom, D. D.; Maurer, P. C. A Fluorescent-Protein Spin Qubit. arXiv, 2024.
(38) Abrahams, G.; Spreng, V.; Štuhec, A.; Kempf, I.; James, J.; Sechkar, K.; Stacey, S.; Trelles-Fernandez, V.; Antill, L. M.; Ingaramo, M.; York, A.; Tetienne, J.-P.; Timmel, C. R.; Steel, H. Quantum Spin Resonance in Engineered Magneto-Sensitive Fluorescent Proteins Enables Multi-Modal Sensing in Living Cells. bioRxiv, **2024**.
(39) Burd, S. C.; Bagheri, N.; Ingaramo, M.; Condon, A. F.; Mondal, S.; Dowlatshahi, D. P.; Summers, J. A.; Mukherjee, S.; York, A. G.; Wakatsuki, S.; Boxer, S. G.; Kasevich, M. Magnetic Resonance Control of Reaction Yields through Genetically-Encoded Protein:-Flavin Spin-Correlated Radicals in a Live Animal. bioRxiv, 2025.
(40) Meng, K.; Nie, L.; Berger, J.; von Grafenstein, N. R.; Einholz, C.; Rizzato, R.; Schleicher, E.; Bucher, D. B. Optically Detected and Radio Wave-Controlled Spin Chemistry in Cryptochrome. arXiv, 2025.
(41) Yang, N.; Cohen, A. E. Optical Imaging through Scattering Media via Magnetically Modulated Fluorescence. *Opt. Express* **2011**, *19*, 25461.
(42) Evans, E. W.; Dodson, C. A.; Maeda, K.; Biskup, T.; Wedge, C. J.; Timmel, C. R. Magnetic Field Effects in Flavoproteins and Related Systems. *Interface Focus.* **2013**, *3* (5), 20130037.
(43) Bogdanov, A. M.; Mishin, A. S.; Yampolsky, I. V.; Belousov, V. V.; Chudakov, D. M.; Subach, F. V.; Verkhusha, V. V.; Lukyanov, S.; Lukyanov, K. A. Green Fluorescent Proteins Are Light-Induced Electron Donors. *Nat. Chem. Biol.* **2009**, *5* (7), 459−461.
(44) Bogdanov, A. M.; Acharya, A.; Titelmayer, A. V.; Mamontova, A. V.; Bravaya, K. B.; Kolomeisky, A. B.; Lukyanov, K. A.; Krylov, A. I. Turning On and Off Photoinduced Electron Transfer in Fluorescent Proteins by π-Stacking, Halide Binding, and Tyr145 Mutations. *J. Am. Chem. Soc.* **2016**, *138* (14), 4807−4817.
(45) Elowitz, M. B.; Surette, M. G.; Wolf, P.-E.; Stock, J.; Leibler, S. Photoactivation Turns Green Fluorescent Protein Red. *Curr. Biol.* **1997**, *7* (10), 809−812.
(46) Sawin, K. E.; Nurse, P. Photoactivation of Green Fluorescent Protein. *Curr. Biol.* **1997**, *7* (10), R606−R607.
(47) Bassetto, M.; Reichl, T.; Kobylkov, D.; Kattnig, D. R.; Winklhofer, M.; Hore, P. J.; Mouritsen, H. No Evidence for Magnetic Field Effects on the Behaviour of Drosophila. *Nature* **2023**, *620* (7974), 595−599.
(48) Kyriacou, C. P. Magnetic Field Responses in Drosophila. *Nature* **2024**, *629* (8010), E3−E5.
(49) Aumonier, S.; Dupuy, J.; Royant, A. *Structure of the Red Fluorescent Protein mScarlet3 at pH 7.5*; ANR, 2022.
(50) Sehnal, D.; Bittrich, S.; Deshpande, M.; Svobodová, R.; Berka, K.; Bazgier, V.; Velankar, S.; Burley, S. K.; Koča, J.; Rose, A. S. Mol* Viewer: Modern Web App for 3D Visualization and Analysis of Large Biomolecular Structures. *Nucleic Acids Res.* **2021**, *49* (W1), W431−W437.
(51) Berman, H. M. The Protein Data Bank. *Nucleic Acids Res.* **2000**, *28* (1), 235−242.



https://doi.org/10.1021/jacs.5c03997
J. Am. Chem. Soc. 2025, 147, 18088−18099